



Parsa Ghafour

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Tehran, Iran

DoB: 2000-08-31

Marital Status: Single

Military Service: Not Served

Education

PhD in Gravity and Cosmology

Branch: Computational Cosmology

Institute/University: Shahid Beheshti University

Tehran, Iran

GPA: --

September 2025 - Present

Master of Gravity and Cosmology

Branch: Computational Cosmology

Institute/University: Kharazmi University

Tehran, Iran

GPA: 18.79/20

September 2022 -

September 2025

Master of Astronomy and Astrophysics

Branch: Extragalactic Astronomy

Institute/University: Kharazmi University

Tehran, Iran

GPA: 18.64/20

September 2023 -

September 2025

Bachelor of Chemical Engineering

Branch: Food Industry

Institute/University: Islamic Azad University Science and Research Branch

Tehran, Iran

GPA: 17.87/20

September 2018 - June

2022

Research

GraphShed: a parameter-free Graph-based waterShed group finder

Publisher: Submitted to The Astrophysical Journal (Apj)

Link: https://www.researchgate.net/publication/405878389_GraphShed_a_parameter-free_Graph-based_waterShed_group_finder

Authors: Ghafour, Parsa; Ansarifard, Saeed; Jalali Kanafi, Mohammad Hossein; Movahed, Seyed Mohammad Sadegh

June 2026

Cosmic environment as the primary driver of dwarf satellite statistics

Publisher: Astronomy & Astrophysics journal (A&A)

Link: https://www.researchgate.net/publication/404708313_Cosmic_environment_as_the_primary_driver_of_dwarf_satellite_statistics

Authors: Tavasoli, Saeed; Ghafour, Parsa

May 2026

The Filament Rift: Λ CDM's Structural Challenge Against Observation

Publisher: The Astrophysical Journal (Apj)

Link: https://www.researchgate.net/publication/398042870_The_Filament_Rift_LCDM's_Structural_Challenge_against_Observation

Authors: Tavasoli, Saeed; Ghafour, Parsa

November 2025

GrAviPaSt's Lens to the Past: Unveiling the Evolution of Filamentary Structures

Publisher: The Astrophysical Journal (Apj)

Link: https://www.researchgate.net/publication/395240687_GrAviPaSt's_Lens_to_the_Past_Unveiling_the_Evolution_of_Filamentary_Structures

Authors: Ghafour, Parsa; Tavasoli, Saeed

September 2025

- VEGA: Voids idEntification using Genetic Algorithm** December 2025
 Publisher: Journal of Cosmology and Astroparticle Physics (JCAP)
 Link: https://www.researchgate.net/publication/398197327_VEGA_Voids_idEntification_using_Genetic_Algorithm
 Authors: Ghafour, Parsa; Tavasoli, Saeed; Shojaei, Mohammad Reza
- From Voids to Clusters: Mergers and Evolutionary Pathways of Star-Forming and Quenched Low-Mass Galaxies** January 2025
 Publisher: Submitted to Astronomy & Astrophysics journal (A&A)
 Link: https://www.researchgate.net/publication/392530027_From_Voids_to_Clusters_Mergers_and_Evolutionary_Pathways_of_Star-Forming_and_Quenched_Low-Mass_Galaxies
 Authors: Shojaei, Mohammad Reza; Tavasoli, Saeed; Ghafour, Parsa

- ### Teaching
- Teaching Assistant(Computational Physics - Dr. Tavasoli)** February 2024 - June 2024
 Kharazmi University of Tehran
 - Teaching Assistant(Computer Programming - Dr. Khoeini)** February 2024 - June 2024
 Kharazmi University of Tehran

- ### Honors
- Best Poster Presentation - 7th Iranian Computational Physics Conference - PSI** February 2026
 - Top-Ranked M.Sc. Entrance – Gravity and Cosmology - Kharazmi University** May 2024
 - Top-Ranked M.Sc. Entrance – Astronomy and Astrophysics - Kharazmi University** May 2025

- ### Experience
- Research Assistant** January 2025 - August 2025
 Kharazmi University of Tehran
 Tehran, Iran
 - Master's Student** January 2023 - January 2025
 Kharazmi University
 Tehran, Iran

- ### Conferences and seminars
- Unveiling the Evolution of Filamentary Structures Over the Past 8 Gyr Based on Λ CDM Cosmology - Sharif University of Technology** May 18, 2025
 - An investigation into the integrated emission spectra properties of galaxies along the Hubble sequence - Kharazmi University of Tehran** December 8, 2024

Skills

Computational Cosmology

Data Analysis with Python

Computational Python

Machine Learning

Parallel Programming

Artificial Neural Networks

C++

Google Data Studio

GNU Octave

The MathWorks MATLAB

IBM SPSS Statistics

ICDL

Command Prompt Scripting

- ### Certificates
- Cosmological Data Analysis at a glance** December 2023
 Institute: Kharazmi University of Tehran
 - Data Science** September 2023
 Institute: MindLuster E-learning
 - Data Preprocessing** August 2023
 Institute: Great Learning

● Introduction to Machine Learning Institute: Great Learning	September 2023
● About The Large Scale Universe Institute: Iran University of Science and Technology	September 2023
● Advanced Parallel Programming Institute: Kharazmi University of Tehran	December 2023
● Introduction to Data Science Institute: Great Learning	September 2023
● Python Clean Coding Institute: MindLuster E-learning	September 2023
● Quantum Computing Based on Superconducting Circuits Institute: Kharazmi University of Tehran	February 2024
● MATLAB Certified Institute: Shahid Beheshti University	July 2023
● Google Data Studio Institute: MindLuster E-learning	September 2023
● Quantum computer programming w Qiskit Institute: MindLuster E-learning	August 2023

Volunteer Experience

● Board Member - Kharazmi University Astronomical Society	September 2023 - June 2024
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Master's courses

- **Advanced Electrodynamics**
- **Observational Astrophysics and Cosmology**
- **Advanced Computational Physics**
- **Gravity 1**
- **Advanced Astrophysics 1**
- **Advanced Astrophysics 2**
- **Advanced Quantum Mechanics 1**
- **Advanced Statistical Mechanics 1**
- **Cosmology 1**
- **Evolution of Galaxies**
- **Dark Energy and Dark Matter**

References

- **Prof. S.M.S. Movahed**
Professor, Faculty of Physics, Shahid Beheshti University of Tehran
m.s.movahed@ipm.ir
- **Dr. S. Tavasoli**
Assistant Professor, Faculty of Physics, Kharazmi University of Tehran
tavasoli@ipm.ir
- **Dr. S. Khosravi**
Associate Professor, Faculty of Physics, Kharazmi University of Tehran
khosravi.sh@gmail.com
- **Dr. A. Vahedi**
Assistant Professor, Faculty of Physics, Kharazmi University of Tehran
vahedi@ipm.ir

Language

English : Proficient

Social Network

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